```
> restart;#The photon mass
> gram:=5.6*10^32;#1 gram is 5.6*10^32 eV/c^2;
```

$$gram := 5.600000000 \times 10^{32} \tag{1}$$

```
> c := 299792458;
```

$$c := 299792458 \tag{2}$$

```
> gram^(-1);
```

$$1.785714286 \times 10^{-33} \tag{3}$$

```
> 10^(-9)/(5.6*10^32)
```

$$1.785714286 \times 10^{-42} \tag{4}$$

```
> restart; c := 299792458; ec := 1.602176000*10^(-19); Digits := 28;
```

$$c := 299792458$$

$$ec := 1.602176000 \times 10^{-19}$$

$$Digits := 28 \tag{5}$$

```
> ec;
```

$$1.602176000 \times 10^{-19} \tag{6}$$

```
> #photon rest mass in J of wave length 555 nm = 2.5669679·10^(-38) J
    but The rest mass don't varies much with frequency
> solve((0.00678)/((1-V^2/c^2)^(1/2)) = 0.08, V); #The magnetron
```

$$-2.98713878735712876861526856 8 \times 10^{8}, 2.98713878735712876861526856 8 \times 10^{8} \tag{7}$$

```
> V := 2.941137856159333281609533114·10^8;
```

$$V := 2.941137856159333281609533114 \times 10^{8} \tag{8}$$

```
> 1-V/c;# The deley of photnos of 555nm with respect to c;
```

$$0.018942012157179323834246319 2 \tag{9}$$

```
> 1239.84/2; 1239.84/12.64
```

$$619.92000000000000000000000$$

$$98.088607594936708860759493 67 \tag{10}$$

```
> (6.78·10^(-3))/1239.84·10^3;
    #the shift in anngstrom of wavelenght between photons 400nm
    and 650nm thus arund ~0.0054 Angstrom that sould be detected
```

from a Galaxy 100 million LY away

$$0.00546844754161827332559039876 1 \tag{11}$$

> *#Cepheid variables must be used to measure distances smaller than 70-100 million LY to other Galaxy thus the redshift*

> *#Thus the photon rest mass should be around ~ 10^{-8} − 10^{-9} eV*

> $\frac{1239.84}{400}$; $\frac{1239.84}{650}$;

$$3.099600000000000000000000000$$

$$1.907446153846153846153846154 \tag{12}$$

> $V0 := solve\left(\frac{10^{-12}}{\left(1-\frac{V0^2}{c^2}\right)^{\frac{1}{2}}} = 3.0996, V0\right)$;

$$V0 := -2.997924579999999999999999844 \times 10^8, 2.997924579999999999999999844 \times 10^8 \tag{13}$$

> $V1 := solve\left(\frac{10^{-12}}{\left(1-\frac{V1^2}{c^2}\right)^{\frac{1}{2}}} = 1.907446, V1\right)$;

$$V1 := -2.997924579999999999999999588 \times 10^8, 2.997924579999999999999999588 \times 10^8 \tag{14}$$

> $\frac{V0}{V1}$; *E0* := 3.0996; *E1* := 1.907446;

$$\frac{-2.997924579999999999999999844 \times 10^8, 2.997924579999999999999999844 \times 10^8}{-2.997924579999999999999999588 \times 10^8, 2.997924579999999999999999588 \times 10^8}$$

$$E0 := 3.0996$$

$$E1 := 1.907446 \tag{15}$$

> $V0 := \left(1-\left(3\cdot 10^{-9}\cdot E0^{-1}\right)^2\right)^{0.5}\cdot c$; $\frac{V0}{c\cdot 4\cdot 10^{-8}}$;

#Difference at 17 decimal place

$$V0 := 2.997924579999999998595822815 \times 10^8$$

$$2.499999999999999998829042270 \times 10^7 \tag{16}$$

> $V1 := \left(1-\left(3\cdot 10^{-9}\cdot E1^{-1}\right)^2\right)^{0.5}\cdot c$; $\frac{V1}{c\cdot 4\cdot 10^{-8}}$;

#Difference at 17 decimal place

$$V1 := 2.997924579999999996292094024 \times 10^8$$

(17)

$$2.499999999999999996907939245 \times 10^{7} \quad \textbf{(17)}$$

```
> V0/c;#Difference at 18 decimal place
```

$$0.999999999999999999953161690 78 \quad \textbf{(18)}$$

```
> V1/c; #Difference at 17 decimal place
```

$$0.99999999999999999876317569 80 \quad \textbf{(19)}$$